

coverage_ledger

Helix OTA — §11.4.25 Coverage Ledger (1.0.0-MVP)

Revision: 2 **Last modified:** 2026-06-10T14:50:00Z **Authority:** Constitution §11.4.25 (Full-Automation-Coverage) + §11.4.6 (No-guessing) + §11.4.27 (100% test-type coverage) **Scope:** Go control plane (server/) + owned device-side / protocol bricks (submodules/ota-*, submodules/http3) + the Tier-0 device emulator (server/internal/deviceemu) + the emulation test-fabric (docs/design/emulation_fabric/) **Maintainer:** QA lead

Post-authoring update (2026-06-08, commit 5363511). This ledger was authored from the tree just before the §11.4.85 stress+chaos suite landed. The “Stress” and “Chaos” cells marked MISSING for the operational surface (groups / reads / membership / repo-fault) are now **COVERED** by server/internal/api/resilience_test.go (race-clean; evidence in docs/qa/20260608-stress-chaos/). Performance/NFR + adversarial-security suites were in flight the same session — see their qa dirs once committed.

Rev 2 update (2026-06-10). This session landed, with real captured evidence: (a) per-event telemetry duration_ms + bytes_transferred ingested end-to-end (71be1cd); (b) the failure→recall(forward-fix)→recovery OTA lifecycle, proven both in-process over real HTTP (12087bc) and on podman containers 7/7 (71c3f77); (c) the HelixQA LIVE Challenge bank 10/0 via tools/helixqa/run_bank.sh (d5fcf4a); (d) the emulation test-fabric — research + design (8a03a63, de87e21) with the dev-host **P0 tier shipped** and **P1 AVD-on-HVF boot smoke** landed (5d6b3ae). New rows §1a/§1b/§1c below carry these; the fabric tier status is in §5 (DESIGNED/PENDING tiers are **never** counted as coverage per §11.4.6).

Anti-bluff note (§11.4.6 / §11.4.25). Every COVERED cell below cites the test file (and, where useful, the function) that actually exists in the tree as of this revision. A cell is marked COVERED **only** when a test exercising that feature is present. Advanced test types not yet landed are marked MISSING or “in progress (this session)” — they are **not** claimed as coverage. On-device (RK3588 / Orange Pi 5 Max) validation is the one class blocked on hardware; it is marked BLOCKED-HW everywhere it applies and is **never** counted as automated coverage.

Legend

Mark	Meaning
COVERED	A real test exercising the feature exists; file cited.
PARTIAL	Some paths covered, notable gaps remain; file cited.
MISSING	No test of this type for this feature exists yet.
N-A	Test type does not apply to this feature.
BLOCKED-HW	Requires RK3588 hardware; cannot be automated tonight.
in progress	Being authored this session; not yet landed — do NOT count.

Test-type columns: **U**=unit, **I**=integration (httptest / real-Postgres), **E2E**=live-server end-to-end, **Sec**=security, **Str**=stress, **Ch**=chaos, **Perf**=performance, **Chal**=HelixQA Challenge.

1. Control plane (server/) — feature × test-type matrix

Feature	U
Auth (login / JWT / refresh rotation / RBAC)	COVERED api/handlers_auth_test.go:TestLogin,TestRefreshRotation
Artifact upload + signature verify	COVERED api/handlers_artifact_test.go, handlers_artifact_parser.go
Releases (create / get / monotonicity / list)	COVERED api/handlers_release_test.go
Deployments (all-targets / conflict / unknown-release)	COVERED api/handlers_deployment_test.go
Staged rollout engine (phases / halt-on-breach)	COVERED rollout/store_test.go:TestEngineDrivesPhasesToCompletion
Recall / rollback	COVERED api/handlers_recall_test.go:TestRecallRecordsRollback
Audit trail	COVERED api/handlers_audit_test.go:TestAuditRecordsSuccessful, handlers_widen_test.go:TestAuditSinceUntilFilter
	COVERED api/handlers_telemetry_test.go, handlers_widen_test.go

Feature	U
Telemetry reads + overview	
Telemetry ingest (client)	COVERED api/handlers_client_test.go:TestClientTelemetryInge
Telemetry per-event duration_ms + bytes_transferred (ingest→store→read, end-to-end)	COVERED deviceemu/telemetry_fields_test.go:TestEmulatorTeler
Device register + status	COVERED api/handlers_device_test.go (register / validation / conflict
Device groups + members	COVERED api/handlers_group_test.go:TestGroupCRUDLifecycle,T
Client update decision (offer / 200 / 204 / short-circuit)	COVERED api/handlers_client_test.go (offers-delta / 200-behind / 20
Anti-downgrade guard	COVERED api/handlers_client_antidowngrade_test.go:TestClientUpdateNeverOf
Branches / widen (advanced upload + filter edge cases)	COVERED api/handlers_branches_test.go, handlers_widen_test.g
Delta register / select	COVERED api/handlers_delta_test.go:TestDeltaRegisterAndFind
Persistence seam — in-memory Repository	COVERED store/memory_test.go, store/contract_test.go:TestMem
Persistence seam — pgx/PostgreSQL Repository	COVERED (same contract)
Health / readiness probes	COVERED health/health_test.go, api/handlers_health_test.go
Config loading	COVERED config/config_test.go:TestLoadDefaults,TestLoadOver
Transport — HTTP/3 (QUIC) + H2 fallback	COVERED transport/transport_test.go:TestDualTransportServes
Response compression — Brotli / gzip / identity	COVERED api/middleware_compression_test.go (brotli-negotiate / gzi
NFR / load (latency / RPS / p99)	N-A

Feature	U
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1a. OTA lifecycle + Tier-0 device emulator (server/internal/deviceemu) — landed this session

Feature	U	I (httptest/PG)	E2E
Failure → recall (forward-fix) → recovery OTA lifecycle (in-process, real HTTP)	COVERED deviceemu/recall_recovery_test.go:TestRecallRecoveryE2E (12087bc)	COVERED — drives real api.Server + real deviceemu.Device over httptest, asserts every step against real response data	COVERED the in- (evidence recall)
Same lifecycle on real containers (podman, full fidelity)	N-A	N-A	COVERED proven + per-20260 conta tier1
Recall (forward-fix) lifecycle e2e on a standalone live server	COVERED — drives a real booted server	COVERED	COVERED (88bd)

1b. HelixQA Challenge bank — LIVE mode (tools/helixqa/run_bank.sh)

Feature	Status	Evidence
Full HelixQA bank, LIVE mode against an operator-booted server	COVERED — 10 passed / 0 failed / 0 skipped (d5fcf4a);	docs/qa/20260610T143025Z-helixqa-livebank/

Feature	Status	Evidence
	runner tools/helixqa/ run_bank.sh	run_bank_live.txt (RESULT: PASS, mode=live)

Anti-bluff note (§11.4.6 / §11.4.146): the LIVE bank had a RED run first — docs/qa/20260610T142608Z-helixqa-livebank/run_bank_live.txt shows 2/8 (two run_bank LIVE-mode bugs: <pw> placeholder shadowing + port collision) — then the fix (d5fcf4a) and the GREEN 10/0 confirmation. The 10/0 PASS is the authoritative result; it is cited as COVERED, the 2/8 is the documented RED baseline, not coverage.

1c. HelixQA incorporated as a submodule (§11.4.27)

Feature	Status	Evidence
HelixQA submodule wired at submodules/helixqa	COVERED (incorporation) d488e6e — submodule submodule.submodules/helixqa.url = git@github.com:HelixDevelopment/HelixQA.git	.gitmodules entry; gitlink present

2. Device-side / protocol bricks (submodules/) — feature × test-type matrix

Feature	U	I
Protocol types / enums / payload / validation	COVERED ota-protocol/{enums,payload,types,validate}_test.go (round-trips, invalid-enum-rejected, descriptive errors)	COVERED
Artifact validator (hash / sig / version / target / metadata)	COVERED ota-artifact-validator/validator_test.go (TestValidateHash/Signature/Version/Target/Metadata/FailFast)	COVERED
Rollout engine (cohort hashing / decide / phases / halt / window)	COVERED ota-rollout-engine/{cohort,decide,engine}_test.go (determinism, boundaries, monotonic growth, full progression, halt-idempotent, window-hold, post-boot-abort)	COVERED
Telemetry schema (codec / events / health derivation)	COVERED ota-telemetry-schema/{codec,event,health}_test.go (batch round-trip, invalid-event-reject, health counts/rates/verdict/thresholds)	COVERED health_t

Feature	U	I
HTTP/3 server brick (config / TLS / lifecycle / cross-backend)	COVERED http3/pkg/server/server_test.go (config-validate, forces-TLS1.3+H3-ALPN, start-twice, idempotent-shutdown)	COVERED (real round-trip)
Device delta-apply decision (agent)	COVERED ota-android-agent/.../delta/DeltaApplyDecisionTest.kt (12 @Test: base-version/hash match, malformed gates, ordering, mutation-immunity)	N-A (purely on-device)
Verify-before-apply (agent: hash + signature)	COVERED ota-android-agent/.../verify/VerifyBeforeApplyTest.kt (9 @Test: hash/sig accept+reject, ordering, mutation-immunity)	N-A
Agent poll state machine + jitter	COVERED ota-android-agent/.../poll/PollStateMachineTest.kt (10), JitterTest.kt (6)	N-A
Agent protocol codecs (round-trip)	COVERED ota-android-agent/.../protocol/CodecRoundTripTest.kt (11 @Test)	N-A
update_engine bridge (apply request / status / error / payload props)	COVERED ota-update-engine-bridge/core/src/test/.../{ApplyRequest, EngineStatus, EngineError, PayloadProperties} Test.kt	N-A
A/B update + AVB/dm-verity + auto-rollback (on-device)	BLOCKED-HW	BLOCKED

3. Biggest gaps (most → least critical)

- 1. Stress + Chaos test types are essentially absent project-wide (§11.4.85).**
Across both the control plane and the bricks, the Str and Ch columns are MISSING almost everywhere. There is no sustained-load suite, no concurrent-contention suite, and no fault-injection (process-death / network-fault / resource-exhaustion / state-corruption) suite wired into any gate. The only load-adjacent asset is server/tools/loadtest/main.go, which **measures** but **asserts nothing** and is not part of an automated gate.
- 2. Performance / NFR has no automated assertion gate.** loadtest reports p50/p90/p99/RPS from real round-trips (honest, anti-bluff by design) but deliberately asserts no target. No CI/pre-build step compares measured numbers against stated NFR targets, so a regression in latency or throughput would not fail any gate.

3. **Security coverage is real but narrow — it is RBAC/validation-shaped, not adversarial.** Genuine security invariants ARE covered (the artifact trust-boundary `TestArtifactUploadIgnoresRequestSuppliedPubkey`, the anti-downgrade guard, the HTTP/3 TLS1.3 floor, RBAC-forbidden paths on every mutating route). What is MISSING: a dedicated security test type — JWT tamper/expiry/replay, fuzzing of request bodies beyond `http3/.../fuzz_test.go:FuzzConfigValidate`, injection/DoS, and authz-matrix sweeps. Today these live as side-assertions inside unit/integration tests, not as a first-class Sec suite.

Secondary: **E2E breadth.** The live-server E2E (`challenge_operational.sh`) is strong for auth/device/group/audit/telemetry/rollout-gates but does NOT drive the full signed-artifact → release → deployment → rollout-evaluate pipeline end-to-end (it SKIPS-with-reason at step 13 per §11.4.3). Releases, recall, delta-select, and client-update decision have no live-server E2E.

4. On-device validation (the one hardware-blocked class)

All RK3588 / Orange Pi 5 Max on-device classes — real `update_engine` A/B apply, AVB/dm-verity verification, auto-rollback, and live agent poll→download→verify→apply→report against a running server — are BLOCKED-HW and are **not** counted as automated coverage anywhere above. The agent-side **logic** for those flows (delta decision, verify-before-apply, poll state machine, bridge DTOs) IS unit-covered in `submodules/ota-android-agent` and `submodules/ota-update-engine-bridge`; only the on-hardware execution is blocked. Per §11.4.3 this is the correct SKIP/BLOCKED posture — never a PASS-by-default.

5. Emulation test-fabric tiers (`docs/design/emulation_fabric/`) — status, NOT coverage

The emulation test-fabric is a phased roadmap (`docs/design/emulation_fabric/ROADMAP.md`, research report 8a03a63, design de87e21). Per §11.4.6, **only tiers with real captured evidence are counted as coverage**; DESIGNED / PLANNED tiers are tracked here for honesty but are **never** counted in the matrices above.

Tier / Phase	Status	Evidence (if shipped)
P0 — T0 ota-device-emulator (dev-host)	SHIPPED	full-lifecycle (2391cb6), fleet (f5a3428), recall→recovery in-proc + podman (§1a), telemetry-fields (§1. Telemetry-rich-fields) — all with docs/qa/ transcripts
P1 — T1 Android AVD on Apple HVF	PARTIAL — boot smoke only. AVD boots arm64-v8a on HVF,	COVERED (boot smoke) tests/emulator/tier1_avd_hvf_smoke.sh

Tier / Phase	Status	Evidence (if shipped)
	asserts sys.boot_completed=1 + abi==arm64-v8a (accel proof)	(5d6b3ae); evidence docs/ qa/20260610T144447Z- avd-hvf-smoke/ boot_smoke_result.txt (booted=1 abi=arm64- v8a sdk=35 + Vulkan/ HVF accel lines). PENDING: agent-harness APK on the AVD + GSI-A/ B real-update_engine question (REPORT §2.2) — NOT done, NOT counted
P2 — Tfw firmware/ U-Boot (QEMU)	DESIGNED (planned PWU)	—
P3 — pkg/fabric target registry + scheduler	DESIGNED (planned PWU)	—
P4 — T2 Cuttlefish (Linux-KVM CI)	DESIGNED (planned PWU; honest macOS- M3 KVM gap per §11.4.112)	—
P5 — Tcp control- plane isolation (microVM)	DESIGNED (planned PWU)	—
P6 — distributed control plane (LAVA)	DESIGNED (planned PWU)	—
P7 — T3 real RK3588 HIL	DESIGNED (planned PWU; real hardware, §11.4.133-gated)	—

§11.4.6 posture: P0 is SHIPPED with evidence; P1 is **boot-smoke only** (the AVD boots with acceleration proven — the agent-drive + real-update_engine layer is explicitly PENDING and is NOT claimed). P2–P7 are DESIGNED roadmap PWUs with no captured evidence — they are **not** coverage.